

## Project Executable Files

|                      |                                         |
|----------------------|-----------------------------------------|
| <b>Date</b>          | 04 July 2026                            |
| <b>Team ID</b>       |                                         |
| <b>Project Name</b>  | Rising Waters – Flood Prediction System |
| <b>Maximum Marks</b> | 3 Marks                                 |

### Project Executable Files

Submission of all executable and deployable files for the project, organized for the evaluator to run or deploy directly.

#### Step 1: Submission Checklist

| S.NO | Item to Submit                                           | Submitted (Yes / No / NA)                                                                                                                              |
|------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Complete source code (all files and folders)             | Yes                                                                                                                                                    |
| 2    | README / Setup Guide                                     | Yes (README.md + AUTH_SETUP.md)                                                                                                                        |
| 3    | requirements.txt                                         | Yes                                                                                                                                                    |
| 4    | Database schema / seed files                             | Yes — SQLAlchemy models (User, Prediction) in models.py; tables are created automatically via db.create_all() on first run, no manual seed file needed |
| 5    | Environment configuration file (.env.example or similar) | Yes — .env.example lists SECRET_KEY, DATABASE_URL, SESSION_COOKIE_SECURE, and optional Google OAuth credentials                                        |
| 6    | Deployed application URL                                 | Yes <a href="https://risingwaters-1.onrender.com/">https://risingwaters-1.onrender.com/</a>                                                            |
| 7    | APK / Executable binary                                  | NA (web app, not mobile/desktop)                                                                                                                       |
| 8    | Dockerfile / Containerization config                     | No (deployed via Procfile + Gunicorn instead)                                                                                                          |
| 9    | Test files and test results                              | Yes (Postman collection)                                                                                                                               |
| 10   | Demo video or walkthrough                                | Yes (video demo script provided)                                                                                                                       |

## Step 2: File / Folder Structure

Rising-Waters/ app.py, extensions.py, models.py, forms.py, requirements.txt, Procfile, runtime.txt, floods.save, transform.save, architecture.png, README.md, AUTH\_SETUP.md, .env.example | dataset/flood dataset.xlsx | notebooks/Rising\_Waters.ipynb (+ comparison model .pkl files) | templates/home.html, predict.html, chance.html, no\_chance.html, login.html, register.html, history.html, partials/nav.html, partials/flash.html, partials/footer.html, errors/404.html, errors/500.html | static/main.css, main.js, hero-rain.png

## Step 3: Deployment / Access Details

| Field                       | Details                                                                                                                                                                             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hosted / Deployed URL       | <a href="https://risingwaters-1.onrender.com/">https://risingwaters-1.onrender.com/</a>                                                                                             |
| Login Credentials (Demo)    | Registration required — create an account at /register, or sign in with “Continue with Google”; no fixed demo credentials are provided                                              |
| Platform / Hosting Provider | Render                                                                                                                                                                              |
| Repository Link             | <a href="https://github.com/Laharisrikotipalli/RisingWaters">https://github.com/Laharisrikotipalli/RisingWaters</a>                                                                 |
| Demo Video Link             | <a href="https://drive.google.com/file/d/1mrpNulO4c-xLdWXaaDxH9CV9T5jWIXNK/view?usp=sharing">https://drive.google.com/file/d/1mrpNulO4c-xLdWXaaDxH9CV9T5jWIXNK/view?usp=sharing</a> |

## Step 4: Run Instructions

1. git clone <https://github.com/Laharisrikotipalli/RisingWaters.git>
2. cd "Rising-Waters/5. Project Development Phase/Source Code"
3. cp .env.example .env (then set SECRET\_KEY to a random value)
4. pip install -r requirements.txt
5. python app.py (creates floods.db automatically on first run)
6. Open <http://127.0.0.1:5000>, register an account, then use /predict

## Step 5: Known Issues / Limitations

| S.N<br>O | Known Issue / Limitation                                        | Workaround / Status                                                                  |
|----------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1        | No live weather API integration; all values entered manually    | Planned as a future enhancement                                                      |
| 2        | Missing or invalid form fields                                  | Resolved — WTForms now shows inline validation errors instead of a generic Flask 400 |
| 3        | No automated test suite (unit/integration tests)                | Currently tested manually via Postman/curl                                           |
| 4        | No password-reset email flow or rate limiting on login/register | Noted in AUTH_SETUP.md as a next step; not yet implemented                           |
| 5        | SQLite database may not persist across redeloys on some hosts   | Set DATABASE_URL to a Postgres connection string in production instead               |